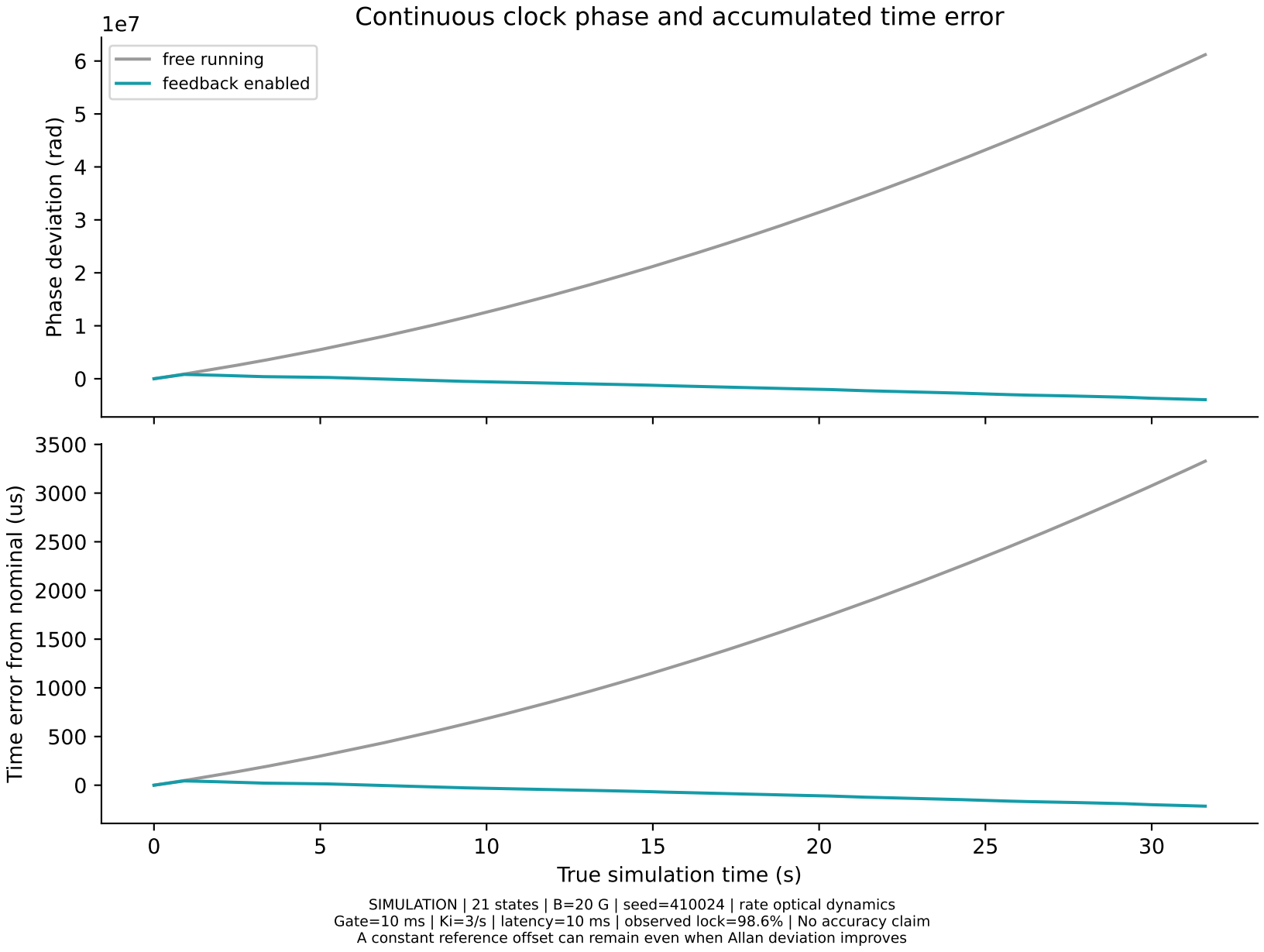
Continuous clock phase and accumulated time error
1e7
free running
feedback enabled
Phase deviation (rad)
6
5
4
3
2
1
0
Time error from nominal (us)
3500
3000
2500
2000
1500
1000
500
0
0
5
10
15
20
25
30
True simulation time (s)
SIMULATION | 21 states | B=20 G | seed=410024 | rate optical dynamics
Gate=10 ms | Ki=3/s | latency=10 ms | observed lock=98.6% | No accuracy claim
A constant reference offset can remain even when Allan deviation improves